

CORE-ET

FROM OPEN ARCHITECTURE
TO OPEN SILICON

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RISC-V opened the ISA; fabricated silicon is still gated by closed IP, validation cost and tool access.

Erbium is the case study: a public commercial-grade edge-AI SoC brought from inherited Esperanto IP to fab submission in 9 months.

CORE-ET ETASP turns that lesson into an agentic RTL workflow: translate compatible IP into clean SystemVerilog, then integrate reusable blocks into systems.

ERBIUM

THE 9-MONTH TAPEOUT

Built from 10 years of Esperanto IP, Erbium is a public commercial-grade edge-AI SoC: manuals, RTL, test benches, build scripts and CI moved into the open while the 8-minion, 16 MB MRAM design was brought up, verified and sent to fab in 9 months.



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SYSTEM SPEC

COMPUTE

- 8
- ET-Minion cores · RV64IMFC
- 16
- hardware threads
- 8
- SIMD lanes · FP / int / tensor

MEMORY

- 16 MB
- embedded MRAM
- 16 KB
- OTP region
- 32 KB
- L1 I-cache · shared
- 4 KB
- L1 D-cache · per core

I/O

- 5
- peripherals · UART I2C QSPI GPIO JTAG
- 1
- xSPI · external host / flash bus

SHIPS OPEN · APACHE 2.0

- 1
- programmer's manual (Erbium TRM)
- 1
- full RTL + reproducible test bench
- ∞
- build scripts + public CI

Open release: RTL, documentation, test benches, build scripts and CI are public.

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METHODOLOGY

1

OPEN THE IP STACK

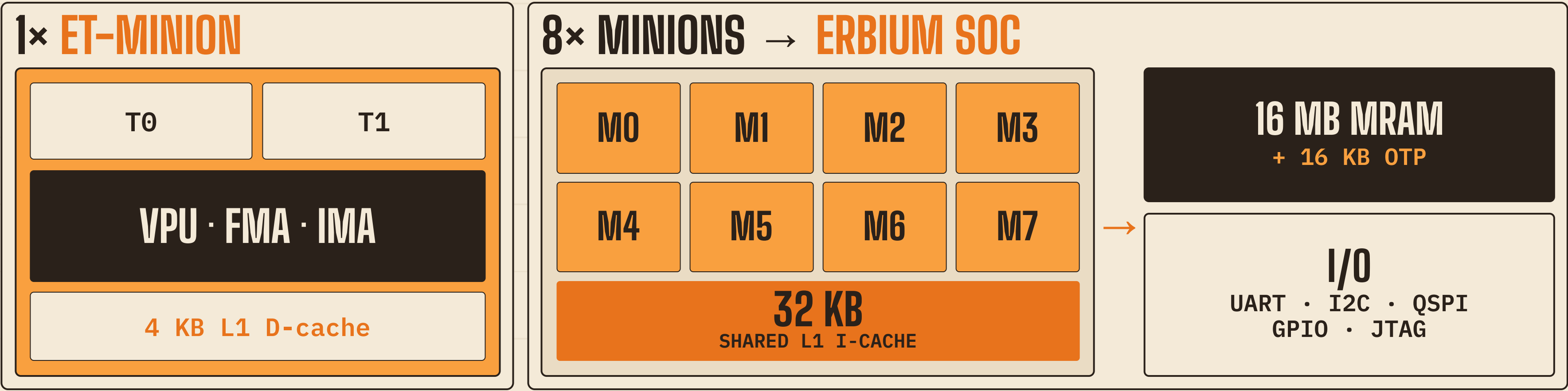
Relicense Esperanto IP under Apache 2.0: toolchain, docs, RTL and verification collateral.

open-sourced under  

2

CONNECT THE BLOCKS

Assemble eight dual-thread ET-Minion cores, shared cache, MRAM and I/O into one coherent Erbium SoC.



3

BOOT · VERIFY · RUN WORKLOADS

BOOT

ZEPHYR

GCC/LLVM and firmware bring-up.

VERIFY

VERILATOR

RTL co-simulation and DV regressions.

RUN WORKLOADS

ML MODELS

Whisper Tiny | Depth Anything V2 | Foundation Stereo | YOLOv5n | YOLOv8n | DnCNN3

4

SEND IT TO FAB

CORE-ET ETASP

AGENTIC RTL DEVELOPMENT

The CORE-ET Agentic Silicon Platform (ETASP) collects hardware IP in a form agents can **translate, verify, document and integrate**. It is seeded by 10 years of Esperanto Technologies IP translated into clean SystemVerilog; the larger goal is a reusable IP catalog for FPGA prototyping and ASIC-oriented implementation.



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AGENTIC WORKFLOW – TRANSLATE, INTEGRATE, BUILD REAL CHIPS!

TRANSLATOR

Takes legally available, license-compatible source IP and rewrites it into **ETASP form**: lowRISC-style SystemVerilog, per-IP documentation, explicit parameters, technology primitives, unit tests, RTL co-simulation and coverage.



INTEGRATOR

Builds systems from existing ETASP components. For translated modules, integration should preserve translated behavior and steer changes toward documented parameters, wrappers or new system RTL, not ad hoc edits to the translated block.

